ASSESSMENT 2 – ANSWERS

Course: Artificial Intelligence **Course Code:** CSA17

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1. Backtracking Search – Doctor Shift Assignment

Constraints: D1 cannot work Night; D2 must work before D3; only one doctor per shift; D3 cannot work Morning; each doctor must receive exactly one shift.

Steps: (1) Assign D1 to Morning. (2) Assign D2 to Afternoon. (3) Assign D3 to Night. Check: D1 is not on Night; D2 works before D3; D3 is not on Morning; each shift has one doctor and each doctor has one shift.

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Result: Morning – D1, Afternoon – D2, Night – D3. All constraints are satisfied.

2. BFS Search – Robot in a 5×5 Grid

Manhattan distance: $h(n) = |x_n - x_G| + |y_n - y_G|$.

Steps: (1) Insert S into the queue. (2) Remove the front node and check whether it is G. (3) If not, examine Up, Down, Left, and Right neighbours. (4) Ignore obstacles and visited cells; add valid neighbours to the queue. (5) Repeat until G is reached and trace back the path.

Result: BFS finds a shortest path in number of moves when all moves have equal cost. The assessment does not show the positions of S, G, or the obstacles, so exact node expansions, queue updates, path, and cost cannot be determined.

3. Autonomous Rescue Robot

a) Effect of constraints: The robot can move Up, Down, Left, or Right and must avoid blocked cells. It can sense only adjacent cells, so its knowledge is incomplete. A normal move costs 1; entering a risky zone costs 3 (1 + 2 additional). It must update its map as it discovers new information and choose a safe, low-cost route.

b) Uniform Cost Search (UCS) approach: (1) Put S in a priority queue with cost 0. (2) Remove the position with the lowest accumulated cost. (3) Observe adjacent cells. (4) Ignore blocked cells; add 1 for a normal move and 3 for a risky-zone entry. (5) Add or update positions with their lowest known cost. (6) Update the map and replan when new obstacles or risks are discovered. (7) Continue until G is reached and return the lowest-cost path.

Cost: Total cost = normal moves + 3 × risky-zone entries.

c) AI concepts: Intelligent agent: senses, processes information, and chooses actions to reach the survivor. **Rationality:** chooses actions that reduce cost while avoiding blocked and unsafe routes. **Environment:** partially observable, sequential, discrete, and dynamic; it may be stochastic if unexpected changes occur.

d) Justification: UCS expands the position with the lowest accumulated cost and handles different movement costs. Because sensing is limited, combining UCS with an online search agent lets the robot update its knowledge and replan when new obstacles or risks are found. It can then seek a least-cost route, assuming non-negative costs.